📱 Flutter Material Page – Basic Structure Note

1️⃣ Import Material Package

Every Flutter app that uses Material Design must import:

```
import 'package:flutter/material.dart';
```

2️⃣ Main Function (Entry Point)

The app starts from `main()` function.

```
void main() {
 runApp(MyApp());
}
```

- `runApp()` loads the main widget of your app.

3️⃣ Create a Class (StatelessWidget)

A Flutter page is usually created using a class.

```
class MyApp extends StatelessWidget {
 @override
 Widget build(BuildContext context) {
   return MaterialApp(
     home: HomePage(),
   );
 }
}
```

- `StatelessWidget` → UI does not change.
- `StatefulWidget` → UI can change (dynamic).

4️⃣ MaterialApp

`MaterialApp` is the root widget for a Material Design app.

```
MaterialApp(
 debugShowCheckedModeBanner: false,
 title: 'My Flutter App',
 home: HomePage(),
);
```

Important Properties:

- `title`
- `home`
- `theme`
- `routes`

5️⃣ Scaffold (Page Structure)

`Scaffold` gives a basic page layout structure.

```
class HomePage extends StatelessWidget {
 @override
 Widget build(BuildContext context) {
   return Scaffold(
     appBar: AppBar(
       title: Text("Home Page"),
     ),
     body: Center(
       child: Text("Hello Flutter!"),
     ),
   );
 }
}
```

6️⃣ Scaffold Structure

Scaffold provides:

- ✅ `appBar`
- ✅ `body`
- ✅ `floatingActionButton`
- ✅ `drawer`
- ✅ `bottomNavigationBar`

Example:

```
Scaffold(
 appBar: AppBar(
   title: Text("My App"),
 ),
 body: Center(
   child: Text("Welcome"),
 ),
 floatingActionButton: FloatingActionButton(
   onPressed: () {
     print("Button Pressed");
   },
   child: Icon(Icons.add),
 ),
);
```

7️⃣ AppBar

Top bar of the screen.

```
AppBar(
 title: Text("App Title"),
 centerTitle: true,
 backgroundColor: Colors.blue,
)
```

AppBar Properties:

- `title`
- `centerTitle`

- `backgroundColor`
- `actions`
- `leading`

Example with action:

```
AppBar(
 title: Text("Dashboard"),
 actions: [
   IconButton(
     icon: Icon(Icons.search),
     onPressed: () {},
   )
 ],
)
```

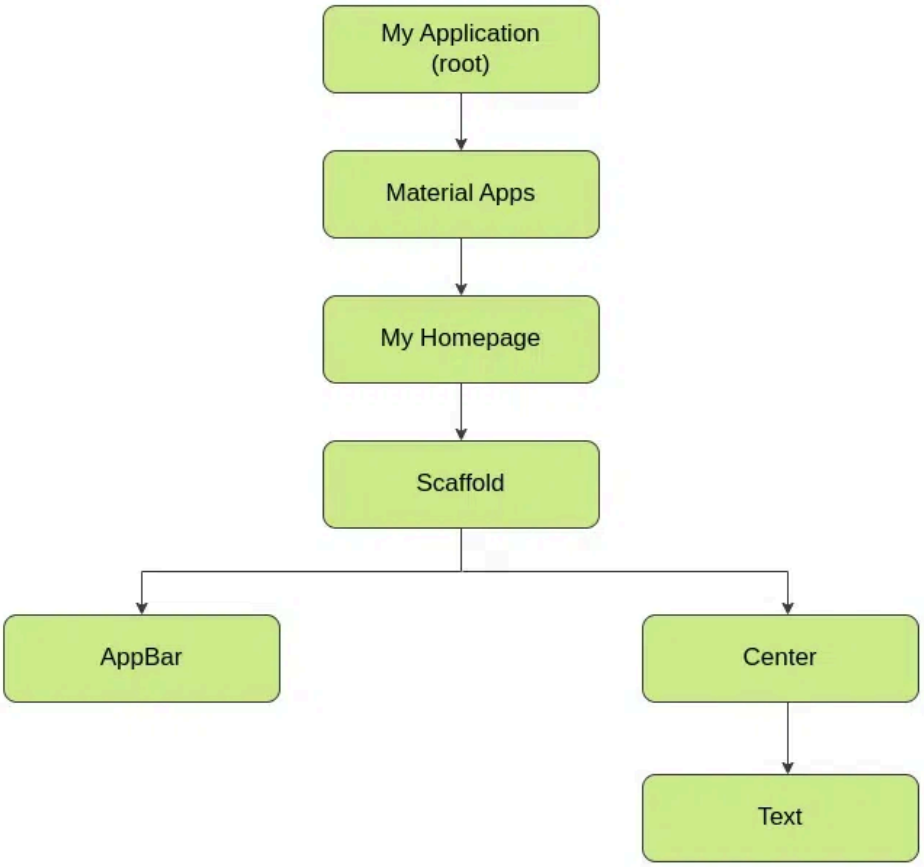
✅ Full Basic Example Code

```
import 'package:flutter/material.dart';

void main() {
 runApp(MyApp());
}

class MyApp extends StatelessWidget {
 @override
 Widget build(BuildContext context) {
   return MaterialApp(
     debugShowCheckedModeBanner: false,
     title: 'Flutter Demo',
     home: HomePage(),
   );
 }
}

class HomePage extends StatelessWidget {
 @override
 Widget build(BuildContext context) {
   return Scaffold(
     appBar: AppBar(
       title: Text("Home Page"),
       centerTitle: true,
```

```
    ),
    body: Center(
      child: Text(
        "Hello Flutter!",
        style: TextStyle(fontSize: 20),
      ),
    ),
    floatingActionButton: FloatingActionButton(
      onPressed: () {
        print("FAB Clicked");
      },
      child: Icon(Icons.add),
    ),
  );
}
}
```

🧠 Summary

- `main()` → Starting point
- `runApp()` → Loads main widget
- `MaterialApp` → Root app widget
- `Scaffold` → Page structure
- `AppBar` → Top navigation bar
- `body` → Main content area



1️⃣ Text Widget

Used to display text.

```
Text(
 "Hello Flutter",
 style: TextStyle(
   fontSize: 20,
   color: Colors.blue,
   fontWeight: FontWeight.bold,
 ),
)
```

Important Properties:

- `style`

- `textAlign`
- `maxLines`
- `overflow`

2️⃣ Container Widget

Used to design and style a box.

```
Container(
 height: 100,
 width: 100,
 padding: EdgeInsets.all(10),
 margin: EdgeInsets.all(10),
 decoration: BoxDecoration(
   color: Colors.blue,
   borderRadius: BorderRadius.circular(10),
 ),
 child: Text("Box"),
)
```

Used for:

- Background color
- Padding & margin
- Border & radius
- Alignment

3️⃣ Row Widget

Arranges widgets horizontally.

```
Row(
 mainAxisAlignment: MainAxisAlignment.spaceEvenly,
 children: [
   Icon(Icons.star),
```

```
  Icon(Icons.favorite),
  Icon(Icons.home),
 ],
)
```

---

4 Column Widget

Arranges widgets vertically.

```
Column(
 mainAxisAlignment: MainAxisAlignment.center,
 children: [
  Text("Item 1"),
  Text("Item 2"),
  Text("Item 3"),
 ],
)
```

---

5 Image Widget

Used to display images.

Asset Image:

```
Image.asset("assets/image.png")
```

Network Image:

```
Image.network("https://example.com/image.png")
```

---

6 Icon Widget

Displays icons.

```
Icon(
 Icons.home,
 size: 30,
 color: Colors.blue,
)
```

---

7 ElevatedButton

Modern button style.

```
ElevatedButton(
 onPressed: () {
   print("Button Clicked");
 },
 child: Text("Click Me"),
)
```

8️⃣ TextField

Used for user input.

```
TextField(
 decoration: InputDecoration(
   labelText: "Enter Name",
   border: OutlineInputBorder(),
 ),
)
```

9️⃣ ListView

Scrollable list of widgets.

```
ListView(
 children: [
   ListTile(title: Text("Item 1")),
   ListTile(title: Text("Item 2")),
   ListTile(title: Text("Item 3")),
 ],
)
```

🔟 Center Widget

Centers its child.

```
Center(
 child: Text("Centered Text"),
)
```

🔥 Most Commonly Used Basic Widgets

- Text
- Container
- Row
- Column
- Image
- Icon
- Button
- TextField
- ListView
- Center

🧠 Short Interview Answer

> Basic widgets in Flutter are the fundamental UI building blocks like Text, Container, Row, Column, Image, Icon, Button, and TextField used to design layouts and user interfaces.

We will use:

- ✅ Scaffold
- ✅ AppBar
- ✅ Column
- ✅ Container
- ✅ Text
- ✅ TextField
- ✅ Icon

- ✅ ElevatedButton
- ✅ SizedBox
- ✅ Center

📱 Simple Flutter Login Page Example

```
import 'package:flutter/material.dart';

void main() {
 runApp(MyApp());
}

class MyApp extends StatelessWidget {
 @override
 Widget build(BuildContext context) {
   return MaterialApp(
     debugShowCheckedModeBanner: false,
     home: LoginPage(),
   );
 }
}

class LoginPage extends StatelessWidget {
 @override
 Widget build(BuildContext context) {
   return Scaffold(
     appBar: AppBar(
       title: Text("Login Page"),
       centerTitle: true,
     ),
     body: Center(
       child: SingleChildScrollView(
         child: Padding(
           padding: EdgeInsets.all(20),
           child: Column(
             mainAxisAlignment: MainAxisAlignment.center,
             children: [

               // Logo Icon
               Icon(
                 Icons.lock,
                 size: 80,
```

```
  color: Colors.blue,
),

SizedBox(height: 20),

// Title
Text(
  "Welcome Back",
  style: TextStyle(
    fontSize: 24,
    fontWeight: FontWeight.bold,
  ),
),

SizedBox(height: 30),

// Email Field
TextField(
  decoration: InputDecoration(
    labelText: "Email",
    prefixIcon: Icon(Icons.email),
    border: OutlineInputBorder(),
  ),
),

SizedBox(height: 20),

// Password Field
TextField(
  obscureText: true,
  decoration: InputDecoration(
    labelText: "Password",
    prefixIcon: Icon(Icons.lock),
    border: OutlineInputBorder(),
  ),
),

SizedBox(height: 30),

// Login Button
Container(
  width: double.infinity,
  height: 50,
  child: ElevatedButton(
    onPressed: () {
      print("Login Button Pressed");
    },
    child: Text(
```

```
            "Login",
            style: TextStyle(fontSize: 18),
          ),
        ),
      ),

      SizedBox(height: 15),

      // Register Text
      Text(
        "Don't have an account? Register",
        style: TextStyle(color: Colors.grey),
      ),

    ],
  ),
),
),
),
);
}
}
```